SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product information

Sales No.	:	UAB12636/00 GINGER ANC IFRA 9999-34-1811 UAB12636/00
Relevant identified uses of the substance or mixture and uses advised against Intended Use Fragrances	:	Perfume compound
Company	:	Givaudan Fragrances Corp. 300 Waterloo Valley Road International Trade Center MOUNT OLIVE NJ 07828 UNITED STATES OF AMERICA +12152070061
Emergency telephone number	:	+18669280789

Please refer to section 16 for a full list of emergency phone numbers.

SECTION 2. HAZARDS IDENTIFICATION

GHS classification in accordance with the OSHA Hazard Communication Standard (29 CFR 1910.1200)

Flammable liquids	:	Category 3
Skin irritation	:	Category 2
Eye irritation	:	Category 2A
Skin sensitisation	:	Category 1
Reproductive toxicity	:	Category 2
Aspiration hazard	:	Category 1

GHS label elements

Hazard pictograms	:	
Signal word	:	Danger
Hazard statements	:	H226 Flammable liquid and vapour. H304 May be fatal if swallowed and enters airways. H315 Causes skin irritation. H317 May cause an allergic skin reaction. H319 Causes serious eye irritation. H361 Suspected of damaging fertility or the unborn child.
Precautionary statements	:	**Prevention:** P201 Obtain special instructions before use. P202 Do not handle until all safety precautions have been read and understood. P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P233 Keep container tightly closed. P240 Ground and bond container and receiving equipment. P241 Use explosion-proof electrical/ ventilating/ lighting equipment. P242 Use non-sparking tools. P243 Take action to prevent static discharges. P261 Avoid breathing mist or vapours. P264 Wash skin thoroughly after handling. P272 Contaminated work clothing must not be allowed out of the workplace. P280 Wear protective gloves/ protective clothing/ eye protection/ face protection/ hearing protection. P280 Wear protective gloves, protective clothing, eye protection and face protection. **Response:** P301 + P310 IF SWALLOWED: Immediately call a POISON CENTER/ doctor. P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower. P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P308 + P313 IF exposed or concerned: Get medical advice/ attention. P331 Do NOT induce vomiting. P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention. P337 + P313 If eye irritation persists: Get medical advice/

attention.
P362 + P364 Take off contaminated clothing and wash it before reuse.
P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

Storage:

P403 + P235 Store in a well-ventilated place. Keep cool.
P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Hazards not Otherwise Classified. : This material may have the potential to be ignited by static electricity during handling or processing. Use proper grounding techniques when handling this product to avoid electrostatic charges. A detailed static evaluation is recommended for liquid handling/processing operation to determine the potential for generating and accumulating static charge. Reference NFPA 77 “Recommended Practice on Static Electricity” for further safe handling practices.
May cause phototoxicity.

Please note: Mixtures have not been tested for health hazards. The health hazard information presented is provided in accordance with US 29 CFR 1910.1200 and is based on the testing of individual components which have been shown to cause or may cause these health effects when tested at higher concentrations or at full strength.

Ingredients with unknown acute toxicity.:
The following percentage of the mixture consists of ingredient(s) with unknown acute toxicity: **0** %

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Hazardous components

Component	CAS-No.	Concentration [%]
(R)-1-methyl-4-(1-methylethenyl)-cyclohexene	5989-27-5	15.00 - 40.00
Terpenes and terpenoids of Orange Oil	68647-72-3	10.00 - 30.00
Orange oil (sweet)	8028-48-6	10.00 - 30.00
Lime oil distilled	8008-26-2	7.00 - 13.00

Administrative information:
Report Information: SDS_US/EN/GHS_SDS_NA_REGION/30
Sales & Distribution Information: MW01/FR/US13/01/0040005923/0102160612/000011
Shipping Order Information: 35,217,040/31,166,724

Bicyclo[3.1.1]heptane, 6,6-dimethyl-2-methylene- (= Beta-pinene)	127-91-3	3.00 - 7.00
3,7-dimethyl-1,6-octadien-3-yl acetate (= linalyl acetate)	115-95-7	1.00 - 5.00
Eucalyptus oil	8000-48-4	1.00 - 5.00
2-methoxy-4-(2-propen-1-yl)-phenol (eugenol)	97-53-0	1.00 - 5.00
3,7-dimethyl-1,6-octadien-3-ol (= Linalool)	78-70-6	1.00 - 5.00
Clove, ext.	8000-34-8	1.00 - 5.00
2,6,6-trimethyl-Bicyclo[3.1.1]hept-2-ene (= alpha-pinene)	80-56-8	0.50 - 1.50
Coriander seed oil	8008-52-4	0.50 - 1.50
3,7-Dimethyl-2,6-Octadienal (Cis & Trans Isomers) (= Citral)	5392-40-5	0.50 - 1.50
2-oxabicyclo(2.2.2)octane, 1,3,3-trimethyl- (= Eucalyptol)	470-82-6	0.10 - 1.00
1-Isopropyl-4-methylcyclohexa-1,4-diene (= gamma terpinene)	99-85-4	0.10 - 1.00
7-methyl-3-methylene-1,6-octadiene (= myrcene)	123-35-3	0.10 - 1.00
3,7-dimethyl-2,6-octadienyl acetate (= geranyl acetate)	105-87-3	0.10 - 1.00

The specific chemical identity and/or exact percentage (concentration) of composition has been withheld as a trade secret.

SECTION 4. FIRST AID MEASURES

First aid measures

Description of first aid measures

General advice : Move out of dangerous area.
Show this safety data sheet to the doctor in attendance.
Symptoms of poisoning may appear several hours later.
Consult physician or Poison Control Center. The National Hotline for US Poison Control Centers is 1.800.222.1222; https://www.poison.org

If inhaled : If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.

In case of skin contact : Immediately seek medical attention if chemical entered ear canal.
If skin irritation persists, call a physician.
If on skin, rinse well with water.

If on clothes, remove clothes.

In case of eye contact : Protect unharmed eye.
Remove contact lenses.
Immediately flush eye(s) with plenty of water.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.

If swallowed : Immediately consult Poison Control Center or physician.
Keep respiratory tract clear.
Do NOT induce vomiting.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.

Most important symptoms and effects, both acute and delayed

May be fatal if swallowed and enters airways.
Causes skin irritation.
May cause an allergic skin reaction.
Causes serious eye irritation.
Suspected of damaging fertility or the unborn child.

Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

SECTION 5. FIREFIGHTING MEASURES

Extinguishing media

Suitable extinguishing media : Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Unsuitable extinguishing media : High volume water jet

Special hazards arising from the substance or mixture

Specific hazards during firefighting : Exposure to decomposition products may be a hazard to health.
Cool closed containers exposed to fire with water spray.
Do not use a solid water stream as it may scatter and spread fire.
Do not allow run-off from fire fighting to enter drains or water courses.

Advice for firefighters

Special protective equipment for firefighters	:	Wear self-contained breathing apparatus for firefighting if necessary.
Further information	:	In the event of fire and/or explosion do not breathe fumes. Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. For safety reasons in case of fire, cans should be stored separately in closed containments. Use a water spray to cool fully closed containers.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

Keep people away from and upwind of spill/leak.
In the case of vapour formation use a respirator with an approved filter.
For emergency conditions, use an approved positive-pressure self-contained breathing apparatus.
Material can create slippery conditions.
Use personal protective equipment.
Ensure adequate ventilation.
Remove all sources of ignition.
Evacuate personnel to safe areas.
Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.

Environmental precautions

Prevent product from entering drains.
If the product contaminates rivers and lakes or drains inform respective authorities.

Methods and materials for containment and cleaning up

Clean contaminated floors and objects thoroughly while observing environmental regulations.
Provide adequate precautions, such as electrical grounding and bonding, or inert atmospheres.
Use only non-sparking tools.
Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).

SECTION 7. HANDLING AND STORAGE

Precautions for safe handling

Advice on safe handling	:	Keep away from heat/ sparks/ open flames/ hot surfaces. No smoking. Avoid formation of aerosol. Do not breathe vapours/dust. Avoid exposure - obtain special instructions before use. Avoid contact with skin and eyes. For personal protection see section 8. Smoking, eating and drinking should be prohibited in the application area. Take precautionary measures against static discharges. Provide sufficient air exchange and/or exhaust in work rooms. Open drum carefully as content may be under pressure. Dispose of rinse water in accordance with local and national regulations.
Advice on protection against fire and explosion	:	Use explosion-proof equipment. Do not spray on a naked flame or any incandescent material. Take necessary action to avoid static electricity discharge (which might cause ignition of organic vapours). Keep away from open flames, hot surfaces and sources of ignition.

Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers	:	Provide adequate precautions, such as electrical grounding and bonding, or inert atmospheres. No smoking. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Observe label precautions. Electrical installations / working materials must comply with the technological safety standards.
Further information on storage conditions	:	Ambient / 10-30℃ (50-85℉) Dry, well ventilated, preferably full, hermetically sealed
Advice on common storage	:	Protect against light.
Other data	:	No decomposition if stored and applied as directed.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Exposure Guidelines

Components	CAS-No. :	Type:	Value :	Basis :	Revision:
(R)-1-methyl-4-(1-methylethenyl)-cyclohexene	5989-27-5	TWA	30 ppm	US WEEL	2016-08-17

Orange oil (sweet)	8028-48-6	TWA	10 mg/m3	NIOSH REL	2013-10-08
Orange oil (sweet)	8028-48-6	TWA	5 mg/m3	NIOSH REL	2013-10-08
Orange oil (sweet)	8028-48-6	PEL	10 mg/m3	CAL PEL	2014-11-26
Orange oil (sweet)	8028-48-6	PEL	5 mg/m3	CAL PEL	2014-11-26
Bicyclo[3.1.1]heptane, 6,6-dimethyl-2-methylene- (= Beta-pinene)	127-91-3	TWA	20 ppm	ACGIH	2014-03-01
	DSEN: Dermal Sensitization				
2,6,6-trimethyl-Bicyclo[3.1.1]hept-2-ene (= alpha-pinene)	80-56-8	TWA	20 ppm	ACGIH	2014-03-01
	DSEN: Dermal Sensitization				
3,7-Dimethyl-2,6-Octadienal (Cis & Trans Isomers) (= Citral)	5392-40-5	TWA	5 ppm	ACGIH	2014-03-01
	DSEN: Dermal Sensitization Skin: Danger of cutaneous absorption				

Exposure controls

Exposure assessment: Exposures are dependent on the product being handled, the potential for chemical release, and any resulting airborne concentrations or dermal contact. Since product handling and release scenarios vary, and no two workplaces are exactly alike, it is recommended that the potential for exposure be assessed prior to the prod-uct's use or introduction. Exposure assessments should be performed by an occupational hygienist, industrial hygienist, or other qualified occupational or environmental health professional. An exposure assessment should be conducted to determine the efficacy of any ventilation and the need for additional PPE. The PPE indicated below are recommendations for worst-case scenario exposures. An exposure assessment will identify more applicable measures to be implemented. EN and ANSI standards are mentioned in the following recommendations, consult equivalent local standards when required.
PPE is always the last resort to avoid exposure. In any case technical and organisational measures have to be explored and used prior to the selection of PPE. The PPE selection is for operators trained to work with chemicals according to good industrial hygiene and safety practice. Operators have to be trained on the use of PPE.

Engineering measures

Use engineering controls to maintain airborne levels below exposure limit requirements or guidelines. If there are no applicable exposure limit requirements or guidelines, use the product only with adequate ventilation.

Personal protective equipment

Eye/face protection	:	Use safety goggles and faceshield tested according to EN 166/ ANSI Z87.1 or equivalent local standard.
Hand protection	:	Use gloves when handling substance in open systems. Inspect gloves prior to use. Train operators for proper use.If only incidental exposure is expected: (work without direct contact to substance) use gloves tested according EN 16523-1/ASTM F739 or equivalent local standard breakthrough times at least 10 minutes, tested for chemicals indicated in chapter 3 of this SDS. Change gloves frequently. If direct skin contact is expected: use gloves tested according to EN 16523-1/ASTM F739 or equivalent local standard, tested for chemicals indicated in chapter 3 of this SDS. Permeation time must exceed contact time.
Other skin protection	:	Wear working clothes covering arms and legs. Use antistatic clothes when flammable liquids are handled. (EN 1149-5; ASTM F1505/ASTM STP1273 or equivalent local standard). The type of protective equipment must be selected according to the concentration and amount of the hazardous substance at the specific workplace. Use apron and sleeve covers or complete chemical suit if exposure is expected.
Respiratory protection	:	Respiratory protection should be worn when workplace exposures exceed exposure limit requirements or guidelines. If there are no applicable exposure limits or guidelines, use an approved respirator where there is a potential for adverse effects, including but not limited to respiratory irritation or odor, or where indicated by the exposure assessment. Selection of air-purifying or positive-pressure supplied-air will depend on the results of the exposure assessment which includes an evaluation of the specific operations and the potential airborne concentrations. For emergency conditions, use an approved positive-pressure self-contained breathing apparatus. In case a risk analysis proved the cartridge respirator as acceptable, use type: ABEK-P3 (EN 14387) OR Combination Multi-gas/P100 (42CFR84.193; ANSI Z88.7 or equivalent local standard) as a backup to engineering controls. In absence of engineering controls, use self-contained breathing apparatus or full face supplied air respirators. Use respirators and components tested and approved under appropriate government standards such as CEN (EU) or

NIOSH 42 CFR 84(US).

Thermal hazards : Wear appropriate thermal protective clothing, when necessary.

Hygiene measures : Remove contaminated clothing and protective equipment before entering eating areas.
Do not eat, drink or smoke during work.
Wash hands any time after handling the product.

Environmental exposure controls

General advice : Prevent product from entering drains.
If the product contaminates rivers and lakes or drains inform respective authorities.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	:	liquid
Form	:	liquid
Color	:	dark yellow to Golden yellow
Taste	:	not determined
Odor	:	Citrus-like, Aldehydic, Spicy
Odour Threshold	:	Not applicable
Flash point	:	124 °F (51 °C) Method: Grabner miniflash closed cup
Lower explosion limit	:	not determined
Upper explosion limit	:	not determined
Flammability	:	Not applicable
Particle size	:	no data available
Particle Size Distribution	:	no data available
Auto-ignition temperature	:	not determined
Decomposition temperature	:	no data available
pH	:	not determined
Melting point	:	not determined
Boiling point	:	not determined
Vapor pressure	:	1.5742 hPa at 68 °F (20 °C) Calculated (100.0 %)
Density	:	858.97 kg/m3 at 68 °F (20 °C)
Bulk density	:	Not applicable
Solubility/qualitative	:	practically insoluble
Partition coefficient: n-octanol/water	:	Not applicable
Viscosity, kinematic	:	no data available
Relative vapour density	:	no data available
Evaporation rate	:	no data available

SECTION 10. STABILITY AND REACTIVITY

Reactivity

No decomposition if stored and applied as directed.

Chemical stability

No decomposition if stored and applied as directed.

Possibility of hazardous reactions

Hazardous reactions	:	No decomposition if stored and applied as directed.Vapours may form explosive mixture with air.

Conditions to avoid

Conditions to avoid	:	Extremes of temperature and direct sunlight.Heat, flames and sparks.

Incompatible materials

Materials to avoid	:	Not applicable

Hazardous decomposition products

Hazardous decomposition products	:	no data available
Thermal decomposition	:	no data available

SECTION 11. TOXICOLOGICAL INFORMATION

Information on toxicological effects

Acute oral toxicity

ATE	:	> 5,000 mg/kg	
(R)-1-methyl-4-(1-methylethenyl)-cyclohexene	:	LD50: 5,600 mg/kg	Species: Mouse
Orange oil (sweet)	:	LD50: > 5,000 mg/kg	Species: Rat
Lime oil distilled	:	LD50: > 5,000 mg/kg	Species: Rat

3,7-dimethyl-1,6-octadien-3-yl acetate (= linalyl acetate)	:	LD50: 13,934 mg/kg	Species: Rat
Eucalyptus oil	:	LD50: 4,440 mg/kg	Species: Rat
2-methoxy-4-(2-propen-1-yl)-phenol (eugenol)	:	LD50: 2,130 mg/kg	Species: Guinea pig
3,7-dimethyl-1,6-octadien-3-ol (= Linalool)	:	LD50: 2,790 mg/kg	Species: Rat
Clove, ext.	:	LD50: 2,650 mg/kg	Species: Rat
2,6,6-trimethyl-Bicyclo[3.1.1]hept-2-ene (= alpha-pinene)	:	LD50: > 500 - 2,000 mg/kg	Species: Rat
2-oxabicyclo(2.2.2)octane, 1,3,3-trimethyl- (= Eucalyptol)	:	LD50: 2,480 mg/kg	Species: Rat
1-Isopropyl-4-methylcyclohexa-1,4-diene (= gamma terpinene)	:	LD50: 3,650 mg/kg	Species: Rat
7-methyl-3-methylene-1,6-octadiene (= myrcene)	:	LD50: > 5,000 mg/kg	Species: Rat
3,7-dimethyl-2,6-octadienyl acetate (= geranyl acetate)	:	LD50: 6,330 mg/kg	Species: Rat

Acute inhalation toxicity : No data is available on the product itself.

Acute dermal toxicity : No data is available on the product itself.

Acute toxicity (other routes of administration) : No data is available on the product itself.

Skin corrosion/irritation : May cause skin irritation and/or dermatitis.

Serious eye damage/eye irritation : Vapours may cause irritation to the eyes, respiratory system and the skin.

Respiratory or skin sensitisation : Causes sensitisation.

Mutagenicity : No data is available on the product itself.

Teratogenicity : No data is available on the product itself.

Reproductive toxicity
Suspected of damaging fertility or the unborn child.

Components:

citral:
Reproductive toxicity - Assessment : Adverse reproductive effect

Carcinogenicity:

IARC : Rating : 2B*****
*Listed as a Group 2B possibly carcinogenic to humans by the International Agency for Research on Cancer (IARC). In our expert opinion, myrcene should not be characterized as possibly carcinogenic to humans.

Components : 7-methyl-3-methylene-1,6-octadiene (= myrcene)

OSHA : Remarks : No component of this product present at levels greater than or equal to 0.1% is on OSHA's list of regulated carcinogens.

NTP : Remarks : No component of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP.

STOT - single exposure : No data is available on the product itself.

STOT - repeated exposure : No data is available on the product itself.

Aspiration hazard : No data is available on the product itself.

Further information : Solvents may degrease the skin.

Please note: Mixtures have not been tested for health hazards. The health hazard information presented is provided in accordance with US 29 CFR 1910.1200 and is based on the testing of individual components which have been shown to cause or may cause these health effects when tested at higher concentrations or at full strength.

SECTION 12. Ecological information

Ecological information is not reported.

SECTION 13. DISPOSAL CONSIDERATIONS

Product	:	Dispose of in accordance with local, state and federal regulations. The product should not be allowed to enter drains, water courses or the soil. Do not contaminate ponds, waterways or ditches with chemical or used container. Send to a licensed waste management company.
Contaminated packaging	:	Do not expose containers to high temperatures such as in hot work processes. Empty remaining contents. Dispose of as unused product. Do not re-use empty containers. Do not burn, or use a cutting torch on, the empty drum.

Dispose of in accordance with local regulations.

SECTION 14. TRANSPORT INFORMATION

14.1 UN number

49 CFR	:	NA 1993
IMDG	:	UN 1197
IATA	:	UN 1197

14.2 UN proper shipping name

49 CFR	: :	Combustible liquid, n.o.s. (Methyl isopropenyl cyclohexene, Orange terpenes and terpenoids)
IMDG	: :	EXTRACTS, LIQUID
IATA	:	EXTRACTS, LIQUID

14.3 Transport hazard class(es)

49 CFR	:	Comb. liq.
IMDG	:	3

Administrative information:
Report Information: SDS_US/EN/GHS_SDS_NA_REGION/30
Sales & Distribution Information: MW01/FR/US13/01/0040005923/0102160612/000011
Shipping Order Information: 35,217,040/31,166,724

IATA : 3

14.4 Packing group

49 CFR : III

IMDG : III

IATA : III

14.5 Environmental hazards

49 CFR
Marine pollutant : yes

IMDG
Marine pollutant : yes

14.6 Special precautions for user

49 CFR
Emergency Response Guidebook Number : 128

IMDG
IMDG Code Segregation Group : None

14.7 Transport in bulk according to IMO instruments

Not applicable for product as supplied.

SECTION 15. REGULATORY INFORMATION

Only relevant information is reported

CERCLA Reportable Quantity

This material does not contain any components with a CERCLA RQ.

TSCA list

No substances are subject to a Significant New Use Rule.

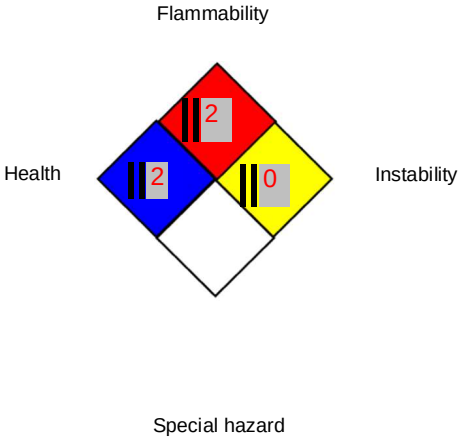
No substances are subject to TSCA 12(b) export notification requirements.

SARA 311/312 Hazards. : Flammable (gases, aerosols, liquids, or solids)
Hazard not otherwise classified (physical hazards)
Respiratory or skin sensitization
Reproductive toxicity
Aspiration hazard
Skin corrosion or irritation
Serious eye damage or eye irritation

SECTION 16. OTHER INFORMATION

Further information

NFPA 704:



Full list of Emergency response numbers worldwide.

	Country	Phone nr		Country	Phone nr
Europe	All Europe	+44 1235 239670	APAC	All East/South East Asia	+65 3158 1074
	France	+33 1 72 11 00 03		Sri Lanka	+65 3158 1195
	Germany	+49 89 220 61012		Taiwan	+886 2 8793 3212
	Spain	+34 91 114 2520		Japan	0120 015 230
	Italy	800 699 792		Indonesia	007 803 011 0293
	Netherlands	+31 10 713 8195		Malaysia	+60 3 6207 4347
	Turkey	0800 621 2139 +44 1235 239670		Thailand	001 800 120 666 751
	Norway	+47 2103 4452		India	+65 3158 1198 000 800 100 7479
	Greece	+30 21 1198 3182		Pakistan	+65 3158 1329
	Portugal	+351 30880 4750		Bangladesh	+65 3158 1200
	Denmark	+45 8988 2286		Philippines	+63 2 8231 2149

Administrative information:
Report Information: SDS_US/EN/GHS_SDS_NA_REGION/30
Sales & Distribution Information: MW01/FR/US13/01/0040005923/0102160612/000011
Shipping Order Information: 35,217,040/31,166,724

	Sweden	+46 8 566 42573		Vietnam	+84 28 4458 2388
	Poland	+48 22 307 3690		Korea	+65 3158 1285
	Czech replublic	+420 228 882 830		South Korea	+82 2 3479 8401
	Finland	+358 9 7479 0199		Australia	+61 2 8014 4558
Middle East/Africa	All Middle East/Africa	+44 1235 239671		New Zealand	+64 9 929 1483
	Bahrain and Middle East	+44 1235 239671		China	+86 532 8388 9090
	Africa/South Africa	+27 21 300 2732	LATAM	Mexico	+52 55 5004 8763
NOAM	USA and Canada	+1 866 928 0789		Brazil	+55 11 3197 5891
	USA and Canada	+1 215 207 0061		Chile	+56 2 2582 9336
	USA and Canada	+1 202 464 2554		Colombia	+57 1 508 7337
Global	Global	+44 1865 407333		Argentina	+54 11 5984 3690

Key or legend to abbreviations and acronyms used in the safety data sheet

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GB EH40 - Great Britain, Environmental Hygiene; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; NOM-010-STPS - Official Mexican Standard, Ministry of Labor and Social Welfare; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of very high concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

Information displayed in section 3 (Composition/information on ingredients) is additional information to understand the hazards of the product and ensure safe handling, storage and transportation. This information, including CAS numbers, is not meant to be used for registration, notification or any other purposes. Any additional information and documentation needed may be provided by Givaudan.

Version : 19 0
Revision Date : 09 JUL 2025
Print Date : 24 SEP 2025